Setting Up Python and Your Development Environment

Course preparation guide

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1 Purpose of this document

This document helps you install Python and set up an editor or IDE *before* the course starts, so that we can focus on learning Python during the workshop. It also explains how to activate the free student version of GitHub Copilot in Visual Studio Code (VS Code) using your Lund University student account.

You do not need to install *all* tools. If you are unsure, install Python and VS Code first; Jupyter, PyCharm and Google Colab are optional alternatives you can try later.

2 Install Python

2.1 Step 1: Check if Python is already installed

a) Open a terminal:

- **Windows**: "Command Prompt" or "Windows Terminal"
- **macOS**: "Terminal"
- **Linux**: your usual terminal emulator

b) Type one of the following commands and press `Enter`:

```
python --version
```

or

```
python3 --version
```

c) If you see something like `Python 3.x.y`, Python is already installed and you can skip to Section 3 (Choose an editor / IDE).

d) If you get an error ("command not found" or similar), continue with the installation steps below.

2.2 Step 2: Install Python on Windows

a) Go to the official download page: https://www.python.org/downloads/.

b) Under "Python Releases for Windows", download the latest **Windows installer (64-bit)**.

c) Run the installer:

- On the first screen, **tick the box** "Add Python 3.x to PATH".

- Click “Install Now” and follow the defaults.

d) When the installer finishes, open “Command Prompt” and run:

```
python --version
```

You should now see your Python 3 version.

2.3 Step 3: Install Python on macOS

a) Go to https://www.python.org/downloads/ and scroll to “Python Releases for macOS”.

b) Download the latest **macOS 64-bit universal2 installer**.

c) Double-click the downloaded `.pkg` file and follow the installer steps (you may need your macOS password). Keep the default choices.

d) After installation, open “Terminal” and run:

```
python3 --version
```

You should see a Python 3 version number.

2.4 Step 4: Install Python on Linux

a) Most Linux distributions include Python 3. Check using:

```
python3 --version
```

b) If you need to install or update Python:

- **Debian/Ubuntu** (and similar):

```
sudo apt update
sudo apt install python3 python3-pip
```

- **Fedora** (and similar):

```
sudo dnf install python3 python3-pip
```

c) Verify with:

```
python3 --version
```

Linux package managers are the recommended way to install Python on Linux.

3 Choose an editor / IDE

You will write Python code in an editor or an Integrated Development Environment (IDE). For this course, we mainly support **VS Code** and **Jupyter Notebooks**, but you may also use **PyCharm** or **Google Colab** if you prefer.

4 Visual Studio Code (VS Code)

4.1 Why VS Code?

VS Code is a fast, lightweight code editor that supports many languages (Python, Java, C#, and more) through extensions. You can install only the features you need, which keeps it lean but powerful.

4.2 Install VS Code

a) Go to https://code.visualstudio.com/download.

b) Download the installer for your system:

- **Windows**: choose the **x64 User Installer** (most common).
- **macOS**:
  - Newer Macs (M1, M2, M3, M4, etc.): choose **Apple silicon**.
  - Older Intel Macs: choose **Intel chip**.
- **Linux**: choose the package format that matches your distribution.

c) Run the installer and accept the default options.

4.3 Install the Python extension in VS Code

a) Open VS Code.

b) Click the **Extensions** icon on the left (or press `Ctrl+Shift+X` / `Cmd+Shift+X`).

c) In the search box, type `Python`.

d) Install the extension called **Python** by **Microsoft** (it has millions of downloads). This adds syntax highlighting, code completion, debugging, Jupyter support, and more for Python in VS Code.

e) When prompted, allow VS Code to install related extensions (like Pylance and the Python Debugger) to improve your experience.

4.4 Create and run a first Python file in VS Code

a) In VS Code, choose `File > Open Folder...` and create/open a folder for your coursework.

b) Create a new file called `hello.py`.

c) Type:

```
print("Hello, world!")
```

d) Save the file.

e) Right-click inside the editor and choose `Run Python File in Terminal` (or use the "Run" button if VS Code shows one). VS Code will open a terminal at the bottom and run your file with Python.

5 Jupyter Notebook (local installation)

Jupyter Notebooks let you mix text, code and output in the same document. We will use them heavily during the workshop.

5.1 Install Jupyter using `pip`

This assumes you already installed Python (Section 2).

a) Open a terminal (Command Prompt / Terminal).

b) Install Jupyter Notebook using:

```
python -m pip install --user notebook
```

or, on some systems:

```
python3 -m pip install --user notebook
```

c) After installation, start Jupyter Notebook with:

```
jupyter notebook
```

d) Your default web browser will open the Jupyter interface. From there, you can create new Python notebooks and run code cells.

5.2 Using Jupyter inside VS Code (optional)

The official Python extension for VS Code also includes Jupyter Notebook support, so you can open `.ipynb` files directly in VS Code and run cells there.

6 PyCharm (optional alternative)

PyCharm is a full-featured Python IDE from JetBrains. The **Community Edition** is free and enough for this course.

a) Go to https://www.jetbrains.com/pycharm/download.

b) Download and install the **Community Edition** for your operating system.

c) When creating a new project, select your installed Python 3 interpreter as the project interpreter.

d) Create a new Python file (e.g. `hello.py`), write `print("Hello, world!")`, and use the green "Run" button to execute it.

7 Google Colab (no installation needed)

Google Colab lets you run Jupyter-like notebooks in a web browser, using a remote Python environment (no local installation required).

a) Open a browser and go to https://colab.research.google.com.

b) Sign in with a Google account.

c) Click `New notebook` to create a new Python notebook.

d) In the first cell, type:

```
print("Hello, world!")
```

e) Press `Shift+Enter` to run the cell and see the output.

8 Anaconda Distribution (optional)

Anaconda is an optional all-in-one distribution that bundles Python, many popular data science packages, the `conda` package manager, Jupyter Notebook and the graphical launcher "Anaconda Navigator" in a single install.You *do not* need Anaconda for this course, but it can be useful if you want an easy way to experiment with data science tools on your own machine.

8.1 When should I use Anaconda?

- If you already installed Python and VS Code and they work, you can skip Anaconda completely.
- If you are new to Python and mainly interested in data science, Anaconda can be a convenient "one installer, many tools" option.
- Anaconda installs its *own* Python and packages, separate from the Python you installed earlier, so it is normal to see different Python versions in the regular terminal and in the "Anaconda Prompt".

8.2 Install Anaconda

a) Open a browser and go to the official Anaconda download page:
https://www.anaconda.com/download.

b) Choose your operating system (Windows, macOS, or Linux) and download the latest **Anaconda Distribution** installer for Python 3.

c) After the download finishes:

- **Windows**: double-click the `.exe` file and follow the installer.
  - Choose "Just Me" (recommended) unless you know you need "All Users".
  - Keep the default installation location.
  - Leave the advanced options at their defaults, then click "Install".
- **macOS**: double-click the `.pkg` file and follow the installer, keeping the default options.
- **Linux**: open a terminal, `cd` to the download folder and run:

```
bash Anaconda3-<VERSION>-Linux-x86_64.sh
```

then follow the prompts and accept the default install location.

d) When the installer finishes:

- On Windows, open the **Anaconda Prompt** from the Start menu.
- On macOS/Linux, open a new terminal window.

e) In that terminal, check that Anaconda's Python works by running:

```
python --version
```

or

```
python3 --version
```

You should now see a Python version provided by Anaconda.

f) To explore the tools that come with Anaconda, you can start **Anaconda Navigator** (from the Start menu on Windows or from your applications menu on macOS) and launch Jupyter Notebook from there.

9 GitHub account and LU Informatics organization

To use GitHub Copilot for free as a student, you need:

- **VS Code installed** on your computer (Section 4).
- A **GitHub account** that is connected to the LU Informatics organization (this is usually done when you accept a GitHub Classroom assignment).

If you have never used GitHub Classroom before, please:

a) Create a GitHub account (if you do not already have one) at https://github.com.

b) Accept this dummy assignment to join the `lu-informatics` organization: https://classroom.github.com/a/CQ0383Fx.

10 Get free access to GitHub Copilot as a student

GitHub Copilot Pro is free for verified students through **GitHub Education** (the GitHub Student Developer Pack).

10.1 Verify as a student (if needed)

If you are not yet verified as a student:

a) Go to https://education.github.com/pack/join.

b) Sign in with your GitHub account.

c) When asked for your email, use your Lund University student email address ending with `@student.lu.se`.

d) When asked to select your university, choose **Lund University** (www.lunduniversity.lu.se).

e) Follow the verification steps (usually by confirming a verification email). Approval can take from a few hours up to a few days.

Once approved, you will have access to the GitHub Student Developer Pack, which currently includes free Copilot Pro for students.

10.2 Enable Copilot in your GitHub account

a) Sign in to GitHub in your browser.

b) Click your profile picture (top-right) and go to `Settings`.

c) In the left menu, find `Copilot` under "Code, planning, and automation".

d) If you are eligible, you will see a page titled **GitHub Copilot Pro** with a button **Get access to GitHub Copilot**. Click it and follow the prompts.

e) Confirm any terms and save the settings.

11 Install and activate GitHub Copilot in VS Code

11.1 Install the GitHub Copilot extension

a) Open VS Code.

b) Go to the **Extensions** view (icon on the left, or `Ctrl+Shift+X` / `Cmd+Shift+X`).

c) In the search box, type `GitHub Copilot`.

d) Install the extension called **GitHub Copilot** (it should have several million downloads and be published by GitHub).

11.2 Sign in and connect VS Code to GitHub

a) After installation, VS Code will usually show a popup asking you to **Sign in to GitHub**. Click it.

b) Your browser will open. Sign in with your GitHub username and password (not your student ID).

c) When asked, click **Authorize Visual-Studio-Code** to allow VS Code to access your GitHub account.

d) Confirm any remaining dialogs so that the browser can open VS Code and complete the sign-in.

11.3 Activate Copilot access

a) Back in VS Code, you may see a message like "No access to GitHub Copilot found" with a button **Sign up for GitHub Copilot**. Click it.

b) Follow the instructions in your browser. Because you are a verified student with the GitHub Student Developer Pack, you should see that you are eligible for free Copilot Pro and can activate it at no cost.

c) If you do *not* see the free option, wait a bit (sometimes benefits appear a few days after verification) and check your Copilot settings page again as described above.

11.4 Test that Copilot works

a) In VS Code, open a Python file (for example `hello.py`).

b) Start typing a comment that describes what you want, for example:

```
# Write a function that returns the square of a number
```

c) After a short moment, you should see a grey suggestion from Copilot. Press `Tab` to accept the suggestion.

If you see suggestions appear as you type, Copilot is active and ready to help you while you code.

12 If you get stuck

If you run into problems while following this guide, try the following:

- Take a screenshot or note the exact error message.
- Bring your laptop (with charger) and your notes to the first session.
- We will reserve time at the beginning of the course to solve setup problems together.

Having Python and at least one working environment (VS Code, Jupyter, PyCharm, Google Colab, or Anaconda) ready will make it much easier for you to follow the workshop from day one.